real baseline image to Module 2

real baseline image to Module 3

Pipeline input: single baseline fundus image
45° color fundus photograph, ≥ mild NPDR, no clinical metadata

Module 1: Severity Staging & Lesion Segmentation (DRG-Net)

Grading
ResNet-50 classifier, trained on FGADR
ICDR grade 0–4 + softmax probabilities

Segmentation
4 U-Nets (ResNet-50 encoder), one per lesion
MA, HE, EX, SE · trained on FGADR + DDR

Lesion Burden Score (LBS)
lesion px / retinal FOV px

Grade label
baseline (start) stage

Lesion mask
union of the 4 lesion masks (1 channel)

LBS + burden stratum
low / medium / high, per-grade tertiles

start stage

baseline mask

grade

LBS stratum

Module 2: Autoregressive Multi-Stage Trajectory Synthesis
StarGAN-style generator adapted from DRForecastGAN · inference cascade

Generator G(x, c)
input: image + lesion mask (4 ch)
+ target stage c (one-hot)
AdaIN target-stage conditioning
weights: best-G.ckpt

Module 1: Segmentation
4 U-Nets re-segment the
synthesized image
→ next-step lesion mask

Module 1: Grading
ResNet-50 consistency check
against the target stage
(logged, does not alter output)

Stage k+1 image + mask
fed back as next input
repeat until PDR (stage 4)

Training only (not used at inference)

Stage 1 pretraining · EyePACS single graded images (40 ep):
stage-classification, cycle + identity L1, Module 1 biomarker-profile
and monotonic-burden losses; WGAN-GP PatchGAN discriminator
Stage 2 fine-tuning · FIRE + LongDR + Tianjin pairs (50 ep):
affine registration aligns follow-up to baseline → L1 + WGAN-GP;
best epoch = lowest L1 on 10 % held-out eyes / patients

Synthesized fundus images
one per stage, up to PDR

Stage label per step
+ Module 1 consistency score

Module 3: Probabilistic Time-to-Progression
runs in parallel with Module 2

EfficientNet-B4 backbone
input: real baseline image only (380 px)
ImageNet init, independent weights
(not shared with Module 1)

Auxiliary embeddings
LBS stratum (3) + baseline grade (5)

Weibull survival head
shape k, scale λ: $S(t) = \exp(-(t/\lambda)^k)$

Training only

Tianjin, fixed 2-year follow-up (30 ep)
current-status (interval-censored)
Weibull NLL; best epoch = lowest
val NLL (best-model.ckpt)

P(progression within 2 years)
+ survival curve S(t)

Complete Forecast Report

Synthesized future-stage fundus images up to PDR
each captioned with its Module 1 grade check

Est. P(further progression within 2 years) per stage
estimates on synthesized images flagged as untested